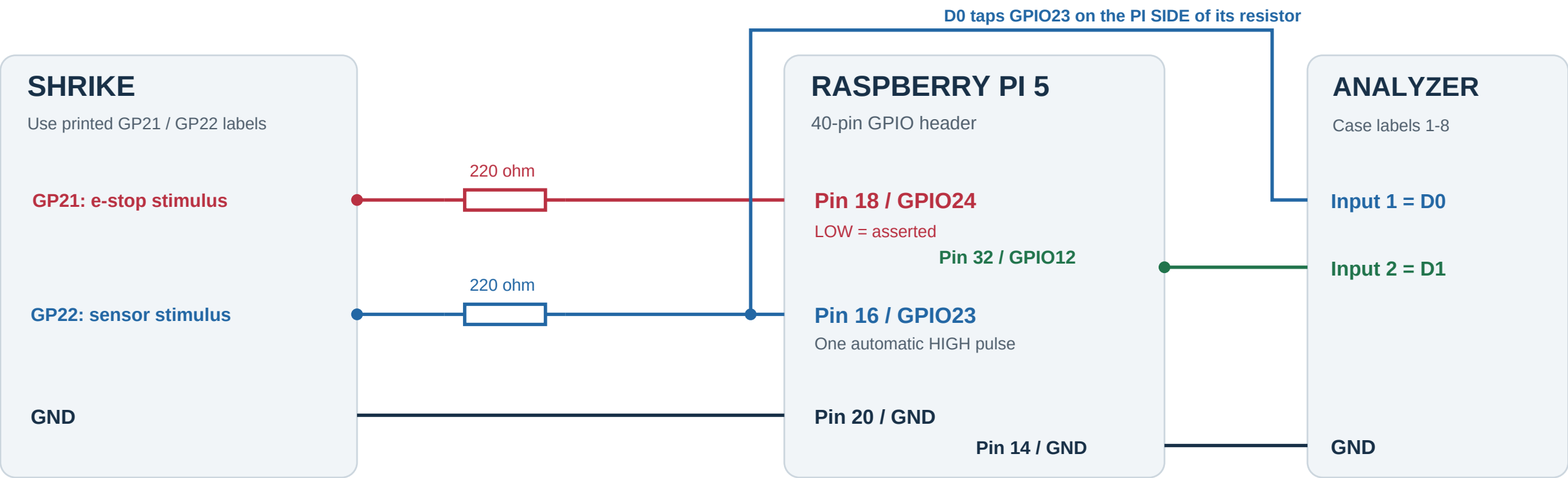


axiomos / PWM carrier + sensor-stop test

Pi 5 + Shrike stimulus + logic analyzer + Debug Probe | No motors, motor drivers or actuators

WIRE WITH BOTH BOARDS UNPOWERED. All GPIO signals are 3.3 V. Do not connect a power rail to GPIO24.

Signal diagram - functional layout, not board orientation. Pi labels are PHYSICAL header pin numbers.



CHANGE FROM E-STOP TEST: input 1 / D0 moves from pin 18 to pin 16. Input 2 / D1 stays on pin 32.

Dots join wires; crossings without dots do not. Pi pins 14 and 20 share ground internally.

USB, UART, SD and power

Host USB -> Shrike USB

Host USB -> Logic analyzer USB

Host USB -> Debug Probe USB -> probe UART port (U) -> Pi dedicated 3-pin UART connector

Keep the existing keyed UART cable unchanged; do not move it to GPIO14/15 on the 40-pin header.

SD card -> Pi microSD slot

Pi USB-C power supply -> Pi power port (OFF until script prompt)

Power-on order

1. Pi power OFF + Shrike USB unplugged: insert the prepared SD and make the wiring above.
2. Connect Shrike USB, analyzer USB and Debug Probe USB. Leave Pi power OFF.
3. Run the PWM capture script. Power the Pi ONLY at "UART RECORDING - POWER ON THE PI NOW".
4. Keep hands off during the automatic sensor pulse. Power off Pi at completion. Do not attach actuators.

Test image: bench-pwm. Requested 10 kHz / 50% PWM; sensor requests duty zero. No automatic rearm. No FPGA in this test.

Source: campaign wiring + scripts/hil/shrike-gpio23-pulse.sh (OUTPUT_MODE=pwm); Pi pin numbering / debug header: official Raspberry Pi documentation.

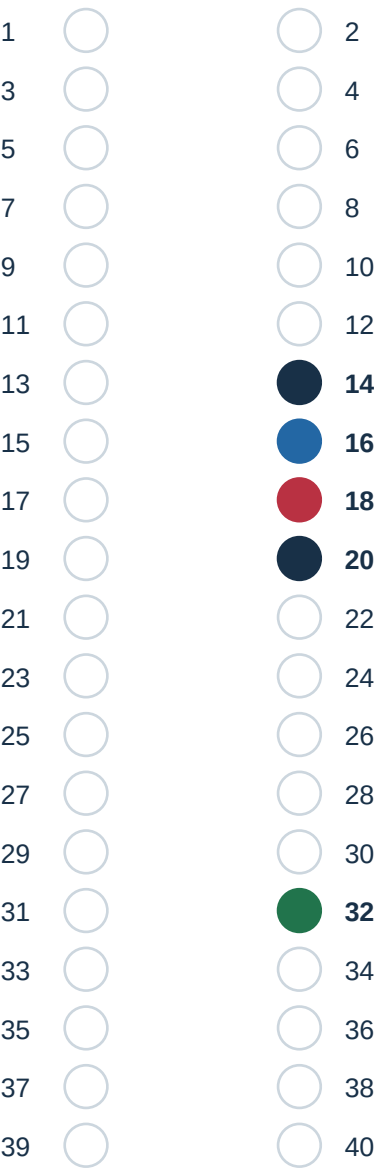
Pin-by-pin check before power-on

Physical pin numbers below refer to the Pi 40-pin header, not BCM GPIO numbers.

Find the actual pin-1 end before counting. This is a numbering map, not an orientation drawing of the board.

Pi header numbering

Pin-1 end at top in this map



Every connection

Shrike GP22

220 ohm resistor -> Pi pin 16 / GPIO23

Analyzer input 1 / D0

Same pin-16 node, after the GP22 resistor

Shrike GP21

220 ohm resistor -> Pi pin 18 / GPIO24

Analyzer input 2 / D1

Pi pin 32 / GPIO12 (PWM output)

Shrike GND

Pi pin 20 / GND

Analyzer GND

Pi pin 14 / GND

Expected capture

D0: LOW -> HIGH -> LOW, one sensor pulse.

D1: PWM carrier before the pulse, then stable LOW after the stop.

Leave all unlisted header pins unconnected for this test. Do not connect Pi and Shrike 3.3 V or 5 V rails.

Use the printed Shrike GP21 / GP22 labels; they are GPIO identities, not Pi-style physical pin numbers.

Retain the existing Debug Probe keyed UART cable to the Pi dedicated 3-pin UART connector.